

Spiking, Recurrent, and Attention Autoencoders for Temporal Signals: A Leakage-Controlled Nested Cross-Validation Comparison

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Abstract—Spiking Neural Networks (SNNs) promise low-power temporal signal processing through event-driven, sparse computation, but their use as autoencoders for waveform reconstruction lacks a controlled, leakage-free empirical comparison against modern non-spiking baselines. We compare four trained autoencoder families — a spiking autoencoder (SNN-AE), an LSTM autoencoder (LSTM-AE), a GRU autoencoder (GRU-AE), and a bottlenecked Transformer autoencoder (Transformer-AE) without encoder–decoder cross-attention — under three spike-encoding front ends (direct, Poisson, latency), together with PCA and mean-frame linear reference points. All models share a single C++/CPU harness [1] for matched timing and a transparent operation-count cost budget. Evaluation uses *nested six-fold leave-one-group-out cross-validation* on three datasets — spoken digits (FSDD [2] and AudioMNIST) and single-lead ECG (MIT-BIH Arrhythmia) — where the group is the speaker or the ECG record: no source recording or group crosses a split, and SNN hyperparameters are selected inside each fold on the held-out validation group only, then retrained on the combined training and validation data before a single evaluation on the untouched test group. Model differences are assessed per dataset with a recording-level paired analysis (bootstrap confidence intervals over recordings, Wilcoxon signed-rank, Holm–Bonferroni), with group-level and seed-level results reported as generalization- and optimization-robustness evidence. We report reconstruction quality (MSE, MAE, R^2) as mean $\pm$ standard deviation over seeds, a raw per-class operation count as the primary cost figure (with a weighted proxy as a labelled sensitivity analysis), the self-attention $O(T^2 d_{\text{model}})$ interaction cost, and real parameter counts. The harness and profiles are released for reproduction.

Index Terms—spiking neural networks, LSTM, GRU, Transformer, autoencoder, temporal signals, surrogate gradients, nested cross-validation, data leakage, audio signal processing

I. INTRODUCTION

One-dimensional temporal signals such as audio waveforms, biosignals, and sensor streams often need to be processed and interpreted in low-power embedded systems where the energy budget is a primary constraint.

Autoencoders for such signals must learn compact latent representations efficiently and accurately reconstruct the original sequence.

Long Short-Term Memory (LSTM) autoencoders are a well-established baseline for sequence reconstruction [3], [4], but their dense matrix operations are computationally expensive. Spiking Neural Networks (SNNs) otherwise communicate via sparse binary events and perform synaptic updates only when

spikes occur, offering potential energy reduction on general-purpose and neuromorphic hardware [5]–[7].

Training SNNs for reconstruction requires backpropagation through non-differentiable spike functions which behave like step functions. Surrogate gradient methods [8], [9] enable end-to-end gradient-based training and have produced competitive SNN classification performance [6], [10]. However, a controlled study of SNN autoencoders for temporal signal reconstruction — compared against several modern non-spiking baselines, under an evaluation protocol that provably prevents information from the test signals leaking into training or hyperparameter selection — is lacking.

Evaluation rigor is the crux. When fixed-length windows from all recordings are pooled, shuffled, and split, windows from the same utterance — and the same speaker — appear in both training and validation, so reported errors partly measure memorization of speaker- and recording-specific waveform detail. Speaker (or patient) identity is the natural unit of generalization, and the publicly available spoken-digit and ECG benchmarks used here have on the order of tens of such groups, which also bounds how strong any population-level claim can be. We therefore evaluate on three datasets and make the recording-level paired difference, not the group-level one, the primary estimand.

This paper addresses these gaps with four contributions:

- 1) A leakage-controlled protocol: *nested six-fold leave-one-group-out cross-validation* on three datasets (FSDD [2], AudioMNIST, MIT-BIH Arrhythmia ECG) in which no source recording or group (speaker / record) crosses a split and SNN hyperparameters are selected on the inner validation group only.
- 2) A comparison of four trained autoencoder families under three input encodings — SNN-AE, LSTM-AE, GRU-AE, and a bottlenecked Transformer-AE without encoder–decoder cross-attention — plus PCA and mean-frame linear reference points, all on one matched C++/CPU harness [1], [11].
- 3) A hierarchical statistical analysis with the recording-level paired difference as the primary estimand (bootstrap over recordings, Wilcoxon, Holm–Bonferroni), and group-level and seed-level results reported as robustness rather than population inference.

- 4) An analytic computational-cost budget reported as a raw per-class operation count (primary) with a weighted proxy as a labelled sensitivity analysis, alongside the self-attention $O(T^2 d_{\text{model}})$ term and real parameter counts; the reproducible framework is released.

II. RELATED WORK

A. Recurrent and attention sequence autoencoders

LSTM autoencoders encode variable-length sequences into fixed-size latent vectors and reconstruct the original sequence from the latent code [3], [4]. The constant-error carousel in the cell recurrence mitigates vanishing gradients during backpropagation through time (BPTT) [12], [13]. The Gated Recurrent Unit (GRU) [14] merges the input and forget gates and drops the separate cell state, giving a smaller recurrent cell with performance comparable to the LSTM on many sequence tasks [15]. Self-attention encoders [16] replace recurrence with a content-based $O(T^2)$ mixing operation and layer normalization [17]. We use all three as trained non-spiking baselines; the Transformer variant is *bottlenecked* — the decoder is conditioned only on the fixed-dimension latent, with no encoder–decoder cross-attention — so that, like PCA and the recurrent autoencoders, all reconstruction information must pass through the latent.

B. Spiking neural networks and surrogate gradients

SNNs model computation via binary spike events governed by membrane potential dynamics [5], [7]. The Leaky Integrate-and-Fire (LIF) neuron is the most widely deployed model. Training with backpropagation requires differentiable surrogates for the discontinuous Heaviside activation; surrogate gradient methods [8], [18], [19] have enabled competitive SNN accuracy with lower inference cost. SNN research has also expanded from classification to temporal sequence modelling and representation learning, but systematic reconstruction benchmarks remain limited [10], [19].

C. Spike encoding for temporal signals

Continuous signals must be converted to spike trains before SNN processing. Rate coding (Poisson spike generation proportional to amplitude), latency coding (earlier spikes for stronger stimuli), and direct pass-through of normalized values represent the three dominant strategies [8], [20]. Encoding choice affects sparsity, gradient flow, training stability, and reconstruction fidelity [8].

D. Evaluation protocol and leakage

On speaker-labelled audio benchmarks, pooling windows before splitting places windows from the same utterance and speaker on both sides of the split, so measured error conflates generalization with memorization. Speaker-disjoint (leave-one-speaker-out) evaluation is the standard remedy; when model or hyperparameter selection is also involved, a *nested* scheme is required so that the test speaker never influences selection or early stopping [21]. The benchmarks here have tens of groups at most, so we treat the group-level result as a robustness check

and make the recording-level paired difference the primary estimand.

III. METHODOLOGY

A. Problem formulation

Let $x \in \mathbb{R}^T$ denote a temporal signal window of length T . An autoencoder learns encoder $E : \mathbb{R}^T \rightarrow \mathbb{R}^d$ and decoder $D : \mathbb{R}^d \rightarrow \mathbb{R}^T$ minimizing reconstruction loss:

$$\mathcal{L} = \|x - D(E(x))\|_2^2. \quad (1)$$

Where x is the input window, $E(x)$ is the latent representation, and $D(E(x))$ is the reconstructed output.

All trained autoencoders use latent dimension $d = 32$; the recurrent models (SNN-AE, LSTM-AE, GRU-AE) use hidden size $H = 64$ and the Transformer-AE uses $d_{\text{model}} = 64$ (Sec. III-D). The window is reshaped into T/f frames of f samples before entering the sequence models. PCA and a mean-frame predictor provide linear reference points, fitted per fold on training data only.

B. LIF neuron model

The continuous-time Leaky Integrate-and-Fire neuron is described by [7]:

$$\tau \frac{dV}{dt} = -(V - V_{\text{rest}}) + RI(t), \quad \tau = RC. \quad (2)$$

Where V is the membrane potential, V_{rest} is the resting potential, $I(t)$ is the input current, R is the membrane resistance, and C is the membrane capacitance. The time constant τ governs the rate of potential decay. The $RI(t)$ term is consistent with the discrete update in Eq. (3).

Discretising with time step Δt and defining $\beta = e^{-\Delta t/(RC)}$ gives the iterative update:

$$V[t] = \beta V[t-1] + RI[t]. \quad (3)$$

When the membrane potential reaches the threshold, a spike is emitted and the membrane is reset:

$$s[t] = \mathbf{1}[V[t] \geq V_{th}], \quad V \leftarrow \begin{cases} 0 & \text{(hard reset)} \\ V - V_{th} & \text{(soft reset)} \end{cases}. \quad (4)$$

Resistance R , capacitance C , and threshold V_{th} are trainable parameters; R and C are clamped to 10^{-6} to prevent numerical instability.

Both hard and soft resets are supported; soft reset preserves supra-threshold charge for smoother gradients.

C. Surrogate gradient training

The spike function $s = \mathbf{1}[V \geq V_{th}]$ has true gradient zero almost everywhere, blocking standard backpropagation [8].

Surrogate gradient methods replace $\partial s / \partial V$ with a smooth approximation in the backward pass while preserving the exact Heaviside rule in the forward pass [8], [18].

Exponential (SuperSpike [9]):

$$\hat{\sigma}'(V) = \frac{1}{\sigma_s} \exp\left(-\frac{|V - V_{th}|}{\sigma_s}\right). \quad (5)$$

Exponential surrogate gradient. σ_s controls the width of the approximation; $\sigma_s = 1$ is used in this work.

Boxcar:

$$\delta'(V) = \mathbf{1}[|V - V_{th}| < \frac{w}{2}]. \quad (6)$$

Boxcar surrogate gradient. w controls the width of the approximation.

We use the exponential surrogate ($\sigma_s = 1$). BPTT [12] unrolls both LSTM and SNN through the full time dimension T ; the reverse-time recurrence for layer l is:

$$\delta_l[t] = \delta'(V[t]) \odot (W_{l+1}^\top \delta_{l+1}[t]) + \beta \delta_l[t+1], \quad (7)$$

where $\delta_l[t] \in \mathbb{R}^{n_l}$ is the error signal at layer l and time t , $W_{l+1} \in \mathbb{R}^{n_{l+1} \times n_l}$ is the feed-forward weight matrix mapping layer- l activations to layer- $(l+1)$ pre-activations (so the forward pass computes $W_{l+1} a_l$), $\odot$ is the element-wise product, $\delta'(V[t])$ is the surrogate derivative of Sec. III-C, and $\beta = \partial V_{\text{post}} / \partial V_{\text{pre}} = e^{-\Delta t / (RC)}$ carries the error one step back in time through the membrane recurrence. The spatial term $W_{l+1}^\top \delta_{l+1}[t]$ and the temporal term $\beta \delta_l[t+1]$ are the SNN analogue of the standard BPTT split.

D. Recurrent and attention baselines

LSTM-AE. Standard gate equations — input, forget, and output gates from a dense pre-activation, with cell-state accumulation $C_t = f_t \odot C_{t-1} + i_t \odot g_t$ [4], [13]. The encoder stacks two LSTM layers ($H = 64$); the final hidden state is projected to the $d = 32$ latent by $\text{Linear}(H \rightarrow d) \rightarrow \tanh$. The decoder applies $\text{Linear}(d \rightarrow H)$, replicates the result over time, runs two stacked LSTM layers, and projects each step back to the frame. States reset between samples.

GRU-AE. Identical topology with the LSTM cell replaced by a GRU cell [14], [15]: reset gate r_t , update gate z_t , and candidate $n_t = \tanh(x_t W_n^\top + r_t \odot (h_{t-1} U_n^\top) + b_n)$, $h_t = (1 - z_t) \odot n_t + z_t \odot h_{t-1}$. Dropping the cell state removes one gate’s worth of parameters per layer; the real counts are reported in the per-dataset summary tables (Table II and following).

Transformer-AE (bottlenecked). The encoder embeds each frame to $d_{\text{model}} = 64$, adds sinusoidal positional encodings, applies N post-norm self-attention blocks [16], [17], mean-pools over time, and projects to the latent through a $\tanh$. The decoder expands the latent, replicates it over T/f positions, adds positional encodings, applies N self-attention blocks, and projects back to frames. There is *no encoder-decoder cross-attention*: this is a deliberate design choice so that reconstruction must pass through the fixed-dimension latent, making the model a strict-compression autoencoder directly comparable to PCA and the recurrent autoencoders. Its parameter count is chosen *near* the recurrent baselines, not matched; the actual value is reported in the results.

Sequence-length cost. Per input window, the recurrent baselines cost $O(TdH)$ operations while self-attention costs $O(T^2 d_{\text{model}})$ for the score and context products plus $O(Td_{\text{model}}^2)$ for the projections. At $T/f = 32$ frames this quadratic interaction term is modest, but it is an architecture-level cost that is independent of the reconstruction-error

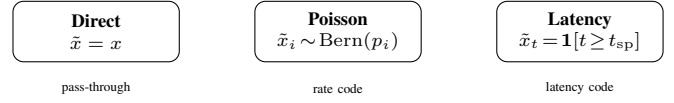


Figure 1. Three spike encoding schemes. Direct passes raw values; Poisson generates stochastic binary spikes proportional to amplitude; latency fires at a time inversely proportional to stimulus strength.

ranking and is reported explicitly in the operation-count budget (Sec. III-I).

E. Spike encodings

Three strategies convert raw window x into pre-processed $\tilde{x}$:

Direct. $\tilde{x} = x$. Normalized values are passed without binarization.

Poisson. Each element is independently binarized via Bernoulli sampling [8]:

$$\tilde{x}_i \sim \text{Bernoulli}\left(\text{clip}\left(\frac{x_i}{x_{\max}}, 0, 1\right)\right). \quad (8)$$

Where $x_{\max}$ is the maximum value in the window after z-score normalization, ensuring the Bernoulli parameter is in $[0, 1]$.

Latency. A binary step fires once per element from a time inversely proportional to signal strength [20]:

$$\tilde{x}_{t,i} = \mathbf{1}\left[t \geq \left\lceil \left(1 - \frac{x_i - x_{\min}}{x_{\max} - x_{\min}}\right) (T - 1) \right\rceil\right]. \quad (9)$$

Where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values in the window after z-score normalization, respectively. Stronger signals fire earlier; weaker signals fire later.

F. SNN pre-processing modes

After spike encoding, a pre-processing transform ϕ is applied before the autoencoder input.

Dense. $\phi(\tilde{x}) = \tilde{x}$. No additional transform.

Conv1d. A fixed 3-tap temporal smoothing filter:

$$\phi(\tilde{x})_t = 0.25 \tilde{x}_{t-1} + 0.5 \tilde{x}_t + 0.25 \tilde{x}_{t+1}, \quad (10)$$

with boundary clamping. Reduces high-frequency spike jitter before autoencoder input.

Recurrent. A recurrent LIF cell integrates the encoded signal:

$$v[t] = \alpha v[t-1] + \tilde{x}[t] - s[t-1] V_{th}, \quad (11)$$

$$s[t] = \mathbf{1}[v[t] \geq V_{th}], \quad (12)$$

where α and V_{th} are sweep parameters.

G. SNN autoencoder architecture

The SNN autoencoder (Fig. 2) is a two-layer fully connected network:

Enc: $\text{Lin}(T \rightarrow H) \rightarrow \text{Leaky} \rightarrow \text{Lin}(H \rightarrow d) \rightarrow \text{Leaky}$

Dec: $\text{Lin}(d \rightarrow H) \rightarrow \text{LeakyInt} \rightarrow \text{Lin}(H \rightarrow T)$

with $T = 256$, $H = 64$, $d = 32$.

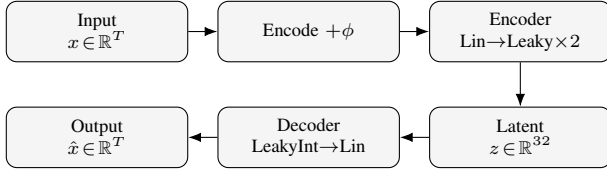


Figure 2. SNN autoencoder architecture. Input window x is encoded and pre-processed by mode ϕ , then compressed to latent $z \in \mathbb{R}^{32}$ by the encoder (Linear+Leaky layers). The decoder (LeakyInt+Linear) reconstructs $\hat{x}$.

Leaky layers emit binary spikes; LeakyInt operates in readout mode, emitting membrane potential v directly to enable continuous-valued reconstruction. The discrete LIF update follows Eq. (3) with $R = 1/V_{th}$ and $C = -1/\ln(\alpha)$ derived from sweep parameters. All membrane states are reset between independent samples.

The four trained families are: SNN-AE (the spiking autoencoder above), LSTM-AE, GRU-AE, and Transformer-AE (Sec. III-D). The spiking pre-processing choices — dense (ϕ identity), conv1d (fixed 3-tap smoothing), recurrent (the LIF input transform above) — are *input transforms* to the one SNN autoencoder, not separate families: the encoder/decoder stack and latent size are fixed and only ϕ changes. Which transform is used for a given fold is chosen by the per-fold selection procedure (Sec. III-D), so SNN-AE denotes the selected variant.

H. Training

All trained models minimize MSE reconstruction loss via Adam [22] ($\eta = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$). SNN biophysical parameters (R , C , V_{th}) use a scaled learning rate $\eta_{bio} = 0.1\eta$ to stabilize membrane-dynamics optimization [8]; R and C are clamped to 10^{-6} in the forward pass with gradients zeroed in the clamped region. BPTT [12] propagates gradients through the full time dimension. Early stopping monitors a held-out monitor MSE with patience $P = 5$ epochs and minimum delta 10^{-8} over a maximum of 30 epochs [10]. Batch size is 1 to accommodate stateful sequence processing. Each configuration is trained with five random seeds (42–46).

Per-fold hyperparameter selection (SNN). Within each outer fold, the full SNN grid — pre-processing mode $\in \{\text{dense, conv1d, recurrent}\}$, threshold $V_{th} \in \{0.5, 1.0, 1.5\}$, membrane decay $\alpha \in \{0.8, 0.9, 0.99\}$, three encodings — is trained on that fold’s four training speakers and scored on the inner validation group. The configuration with the lowest validation MSE is selected, then *retrained* on the training and validation groups combined and evaluated *once* on the held-out test group. Because train and validation groups share no recordings with the test group but not with each other, the retrain’s early-stopping monitor is a recording-disjoint slice carved (seeded, whole recordings) from the training speakers, so no test-speaker signal ever reaches a fitting or stopping decision. The per-fold selected configuration and its inner-validation scores are written to a selection manifest and summarized per dataset in the selection tables (e.g. Table IV).

Fixed-architecture baselines. LSTM-AE, GRU-AE, and Transformer-AE use fixed profile architectures; their only use

of the validation group is early stopping, which never touches the test group. PCA and the mean-frame predictor are fitted per fold on the training speakers only.

I. Analytic computational cost

We report an analytic *computational-cost* figure, not an energy measurement. The primary quantity is the raw operation count

$$C_{\text{raw}} = \sum_k N_k, \quad (13)$$

where N_k is the number of operations of class k executed in one forward pass over a window. We count, separately, the following classes: for the ANN models, dense multiply–accumulates (MACs), nonlinearity evaluations, attention $QK^\top \cdot V$ products, and normalization operations; for the SNN, synaptic accumulations, neuron-state updates, and spike/event operations. The event-driven classes are scaled by the measured mean spike fraction $r \in [0, 1]$ of the evaluated run, so an SNN that never spikes contributes no event-operation cost.

As a secondary, explicitly labelled *sensitivity analysis* we also report a weighted proxy

$$C_{\text{weighted}} = \sum_k w_k N_k, \quad (14)$$

under a stated choice of relative per-class weights w_k ; the ranking of models under C_{weighted} is reported only alongside the weights that produce it, never as a standalone claim.

Neither C_{raw} nor C_{weighted} is energy. Real energy on a given device is dominated by memory movement, cache and SIMD behaviour, clock frequency, supply voltage, exploited sparsity, and the target silicon (CPU, GPU, or neuromorphic). On-device power measurement is future work; the figures here are design-time guidance for comparing architectures under matched software and a single CPU backend.

IV. EXPERIMENTAL SETUP

Datasets.

We evaluate on three datasets, each a corpus of 1-D signals with a natural group unit for leave-one-group-out (Table I):

- **FSDD** [2] — spoken digits (0–9), six speakers, 8 kHz; group = speaker.
- **AudioMNIST** — spoken digits, sixty speakers, offline-resampled from 48 kHz to 8 kHz; group = speaker. The same task as FSDD at ten times the speaker count.
- **MIT-BIH Arrhythmia** — single-lead ECG (lead 0), 48 half-hour records at 360 Hz, converted to physical units from the WFDB header; group = record. A few records share a subject (noted in Sec. VI).

Every recording is partitioned into non-overlapping windows of $T = 256$ samples with stride $S = T$. For FSDD all windows are used; for the two larger datasets a per-recording cap keeps long or silence-padded recordings from dominating the pooled window counts (2 windows/recording for AudioMNIST, 40 for MIT-BIH), which also keeps the three datasets’ training sizes comparable. Adjacent windows of one recording remain

Table I
DATASETS. G = GROUP UNIT; K = OUTER FOLDS; CAP = WINDOWS KEPT PER RECORDING.

Dataset	Signal	G	# G / K	cap
FSDD	digits, 8 kHz	speaker	6 / 6	—
AudioMNIST	digits, 8 kHz	speaker	60 / 6	2
MIT-BIH	ECG, 360 Hz	record	48 / 6	40

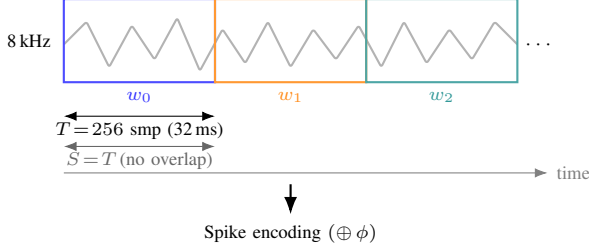


Figure 3. Signal windowing. Each recording is segmented into non-overlapping windows of $T = 256$ samples (32 ms at 8 kHz). Stride $S = T$ yields windows with no shared samples. Windows are assigned to train / validation / test splits by speaker before any pooling.

temporally correlated, which motivates the recording-level statistical unit (Sec. IV-A).

Preprocessing.

Each window is z-score normalised in place (subtract mean, divide by standard deviation) before encoding — a per-window statistic, so it introduces no cross-window leakage. Encodings that require a $[0, 1]$ range (Poisson, latency) re-normalise the z-scored window to its own $(x_{\min}, x_{\max})$ range internally. The three encodings are deterministic transforms (Poisson uses a per-window seeded draw) and are not fitted.

Nested leave-one-group-out cross-validation.

For each dataset the distinct groups are sorted and partitioned into $K = 6$ contiguous blocks. For outer fold $f \in \{0, \dots, 5\}$ the test block is f , the validation block is $(f + 1) \bmod 6$, and the remaining four blocks train. When $\#groups = K$ (FSDD) each block is a single speaker and this is ordinary leave-one-speaker-out; when $\#groups > K$ (AudioMNIST: 10 speakers per block; MIT-BIH: 8 records per block) it is grouped leave-one-block-out. Every window is assigned to a split *by its group, before any pooling*; training windows are shuffled (seeded), validation and test windows kept in (recording, position) order. Each group is in the test block exactly once and the validation block exactly once. At run start the harness asserts that no `recording_id` and no group appears in more than one split and writes a per-fold split manifest (group sets, per-recording window counts); a violation aborts the run. Headline numbers come only from the test split, aggregated across the six folds. SNN hyperparameters are selected on the validation block only, then retrained on $\text{train} \cup \text{validation}$ before a single evaluation on the test block (Sec. III-D).

Evaluation harness.

The C++ harness [1] runs on the XTensor backend [11] (CPU-only, CBLAS, SIMD via xsimd). All models — spiking and non-spiking — are trained and timed on this one backend,

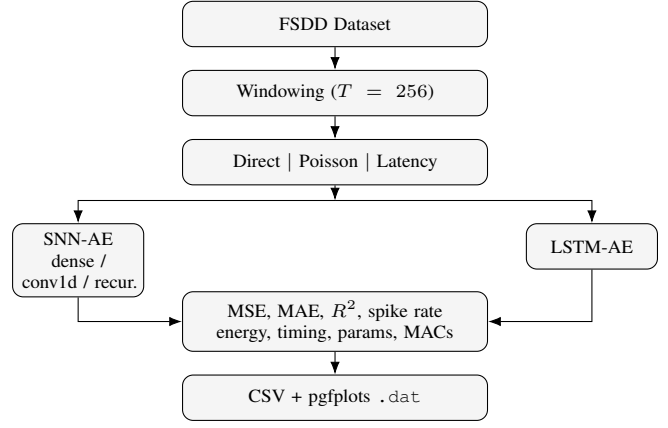


Figure 4. Experiment execution pipeline. Per outer fold, each FSDD window is encoded three ways and processed by the four trained families (SNN-AE with a per-fold selection sweep; LSTM-AE, GRU-AE, Transformer-AE fixed) plus the PCA and mean-frame references. Per-window errors and manifests are written to CSV and JSON.

so the training and inference times in the per-dataset summary tables are directly comparable.

Metrics.

Per run we report reconstruction quality (MSE, MAE, R^2) and, for cost, spike rate (SNN only; 0 for the non-spiking models), the raw per-class operation count of Sec. III-I, parameter count, training time, and inference time in ms. FSDD provides no anomaly labels, so classification metrics are out of scope. Reconstruction error in the headline tables is the mean $\pm$ sample standard deviation over the five seeds (per-seed means first); the inferential analysis is in Sec. IV-A. A per-window error record (model, encoding, fold, split, speaker, recording, window, MSE, MAE) is written for every model so that paired comparisons can be formed at the window, recording, speaker, or seed level.

Profile.

A single JSON profile (`article-losos.json`) configures the harness: the three datasets, six outer folds, five seeds (42–46), the four trained families plus the two linear references, three encodings, and the SNN selection grid ($3 \times 3 \times 3$). The run script iterates over (dataset, fold); a profile audit test verifies the profile parses and validates.

A. Statistical methods

Comparisons are paired on a shared, model-independent window identifier, so every model reconstructs the same windows and differences are formed per window. Let $\text{MSE}_{m,i}$ be the error of model m on window i .

Primary — recording level. For a recording r , let $E_{m,r} = \text{mean}_{i \in r} \text{MSE}_{m,i}$. For model m and reference b the paired difference is $d_r = E_{m,r} - E_{b,r}$. We report $\text{mean}(d_r)$, a 95% confidence interval from a bootstrap that resamples recordings, the Wilcoxon signed-rank p -value on $\{d_r\}$, and Holm–Bonferroni correction across the reference set $b \in \{\text{LSTM-AE}, \text{GRU-AE}, \text{Transformer-AE}, \text{PCA}, \text{Mean}\}$. Effect size is reported both absolute ($\text{mean}(d_r)$) and relative

($\text{mean}(d_r) / \text{mean}(E_{b,\cdot})$). Recordings are the smallest unit across which windows are not, by construction, near-duplicates.

Robustness — group level. Per held-out group (speaker or ECG record) we form the mean error and run a paired Wilcoxon / sign test. This checks whether the sign and rough magnitude of a difference hold across a dataset’s groups; with only tens of groups it is not strong population-level inference and is reported as such.

Descriptive — window level. Paired bootstrap on all windows, reported but explicitly labelled pseudoreplicated (adjacent windows of a recording are correlated), so it is never the basis of a claim.

Robustness — seed level. Paired Wilcoxon across the five per-seed mean errors, characterizing optimization/reproducibility variation around the estimate.

The per-run CSVs are immutable; aggregation and statistics are separate downstream steps that never overwrite them. Synthetic known-answer tests (identical predictions $\Rightarrow p \approx 1$ and CI spanning 0; injected constant offset $\Rightarrow$ significant with CI excluding 0 and recovered offset; label permutation $\Rightarrow$ approximately uniform p) guard the analysis code in continuous integration.

V. RESULTS

All numbers come from the held-out test split only, pooled across the six outer folds, reported as mean $\pm$ sample standard deviation over the five seeds. Each dataset is reported in its own block: a per-model reconstruction and cost table, a per-encoding breakdown, the per-fold SNN selection, the recording-level paired analysis, and an MSE bar chart. The mean-frame predictor and PCA fix the scale of a “no-model” and a “best linear” reference; a trained model is interesting only to the extent it beats PCA, and an ordering among the trained families is asserted only where the Holm-corrected recording-level interval excludes zero. Because SNN selection is redone inside every fold, “SNN-AE” denotes a *selection procedure*; disagreement of the selected mode across folds is itself a finding about how encoding-dependent the best pre-processing is. The cost columns (P , C_{raw} , timing) are architecture properties and do not change with the dataset; the Transformer’s $O(T^2 d_{\text{model}})$ self-attention term shows up in its C_{raw} independently of its MSE.

A. FSDD (spoken digits, 6 speakers)

B. AudioMNIST (spoken digits, sixty speakers)

C. MIT-BIH Arrhythmia (single-lead ECG, 48 records)

Reading across datasets. A model family’s advantage is credible to the extent it recurs on all three datasets’ recording-level analyses; an effect that appears only on FSDD — the smallest — or only on one modality is reported as such. The spoken-digit datasets (FSDD, AudioMNIST) test the same task at two speaker-pool sizes; MIT-BIH tests whether the ranking transfers to a different 1-D signal.

Table II

FSDD (SPOKEN DIGITS, 6 SPEAKERS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD).

P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000			
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0

Table III

FSDD (SPOKEN DIGITS, 6 SPEAKERS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table IV

FSDD (SPOKEN DIGITS, 6 SPEAKERS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD’S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table V

FSDD (SPOKEN DIGITS, 6 SPEAKERS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL — REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{Holm} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{Holm}
—	—	0	0.0000	[0.0000, 0.0000]	1

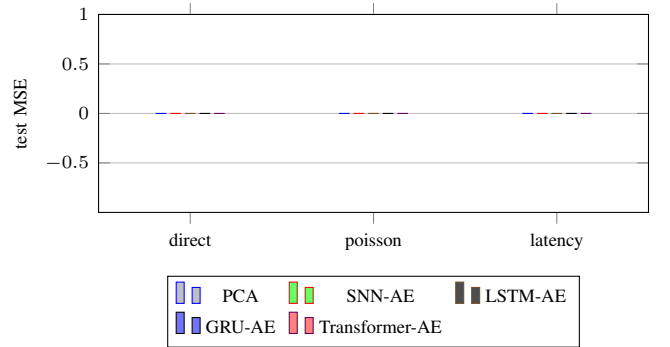


Figure 5. FSDD (spoken digits, 6 speakers): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

VI. DISCUSSION

Two things should be read from the recording-level tables (Table V and following) rather than from raw MSE gaps. First,

Table VI

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}	Model Tr. (ms)	Inf. (ms)	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0

Table VII

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2	Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table VIII

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD'S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table IX

AUDIOMNIST (SPOKEN DIGITS, SIXTY SPEAKERS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL — REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{Holm} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{Holm}
—	—	0	0.0000	[0.0000, 0.0000]	1

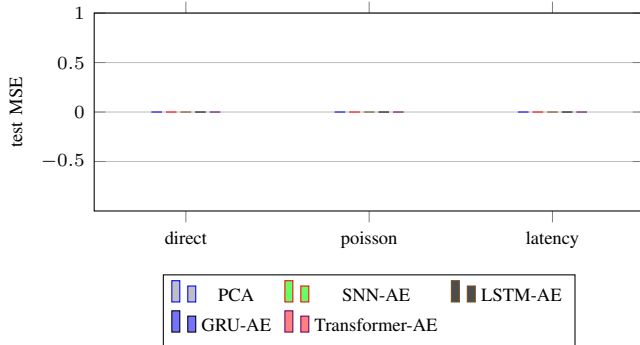


Figure 6. AudioMNIST (spoken digits, sixty speakers): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

whether each trained family beats the PCA reference at all: under speaker-disjoint evaluation, a family whose recording-

Table X

MIT-BIH ARRHYTHMIA (SINGLE-LEAD ECG, 48 RECORDS): TEST RECONSTRUCTION AND COST BY MODEL (MEAN $\pm$ STD OVER FIVE SEEDS; BEST PER COLUMN IN BOLD). P = TRAINABLE PARAMETERS; C_{raw} = RAW PER-FORWARD-PASS OPERATION COUNT. PCA / MEAN ARE ANALYTIC REFERENCES.

Model	MSE	MAE	R^2	P	C_{raw}	Model Tr. (ms)	Inf. (ms)	MSE	MAE	R^2	P	C_{raw}
Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				Mean	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				
PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				PCA	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000				
SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	SNN-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	LSTM-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	GRU-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0
Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	Transformer-AE	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0	0	0

Table XI

MIT-BIH ARRHYTHMIA (SINGLE-LEAD ECG, 48 RECORDS): RECONSTRUCTION PER INPUT ENCODING (MEAN $\pm$ STD OVER FIVE SEEDS).

Model	Enc.	MSE	MAE	R^2	Model	Enc.	MSE	MAE	R^2
SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	SNN-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	LSTM-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	GRU-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	Transformer-AE	direct	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000

Table XII

MIT-BIH ARRHYTHMIA (SINGLE-LEAD ECG, 48 RECORDS): SNN PRE-PROCESSING MODE / V_{th} / α SELECTED ON EACH FOLD'S VALIDATION BLOCK (PARENTHESED FRACTION = SEEDS NOT UNANIMOUS).

Fold	Test grp.	Val grp.	Enc.	Mode	V_{th}	α
0	—	—	—	—	—	—

Table XIII

MIT-BIH ARRHYTHMIA (SINGLE-LEAD ECG, 48 RECORDS): RECORDING-LEVEL PAIRED ANALYSIS. $\bar{d}_r$ = MEAN PER-RECORDING MSE DIFFERENCE (MODEL — REFERENCE); NEGATIVE FAVOURS THE MODEL. CI = 95% BOOTSTRAP OVER RECORDINGS; p_{Holm} = HOLM-CORRECTED WILCOXON SIGNED-RANK.

Model	Reference	n_{rec}	$\bar{d}_r$	95% CI	p_{Holm}
—	—	0	0.0000	[0.0000, 0.0000]	1

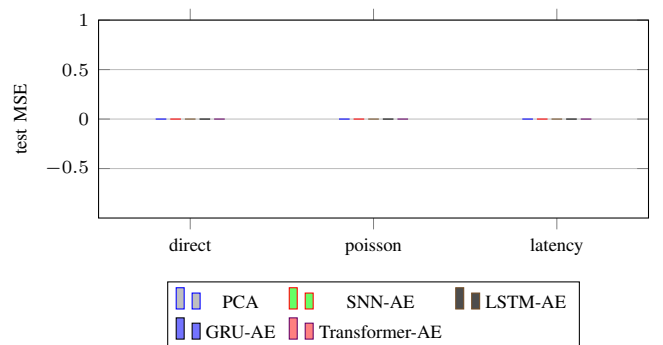


Figure 7. MIT-BIH Arrhythmia (single-lead ECG, 48 records): test MSE by model and input encoding (mean over seeds and folds; mean-frame omitted for scale).

level interval against PCA includes zero has not demonstrated

that its nonlinearity buys anything on this benchmark. Second, the ordering among SNN-AE, LSTM-AE, GRU-AE, and Transformer-AE — a difference is only asserted where the Holm-corrected interval excludes zero. The seed-level and group-level checks in the released JSON say whether an ordering that holds in the pooled recording-level analysis also holds run-to-run and speaker-to-speaker.

The encoding comparison (the per-dataset per-encoding tables) is where an SNN-specific effect is most likely to appear, because the encoding only reshapes the SNN’s input; the non-spiking baselines consume the same framed window regardless. If latency or Poisson coding helps the SNN relative to direct coding, that is a property of spike-based representation, not of the reconstruction task.

The cost picture is separable from the quality picture. The operation-count column of the summary tables orders the models by arithmetic work under matched software; the self-attention $O(T^2 d_{\text{model}})$ term makes the Transformer’s per-window cost grow faster in sequence length than the recurrent models’, independent of which model reconstructs best. Wall-clock timing on the shared CPU backend is reported alongside, but it reflects this implementation and backend, not silicon-level energy.

Limitations.

- **Few groups.** The three datasets have 6, 60, and 48 groups; the group-level analysis is a robustness check, not population inference, which is why the recording-level analysis is the primary estimand.
- **Three small datasets.** Two are the same task (spoken digits) and one is ECG; the ranking may not transfer to other 1-D signal domains. AudioMNIST and MIT-BIH are additionally window-capped per recording (2 and 40) to bound compute, and a few MIT-BIH records share a subject, so its record-level grouping is a slight over-count of independent subjects.
- **Cost is not energy.** C_{raw} and C_{weighted} count operations under matched software; real energy depends on memory movement, sparsity exploitation, and target hardware. On-device measurement is future work.
- **Correlated windows.** Non-overlapping windows of one recording remain correlated, so window-level statistics are pseudoreplicated and used descriptively only.
- **Approximate parameter matching.** The Transformer-AE is sized *near*, not equal to, the recurrent baselines; the summary tables reports the real counts so cost and quality can be read together.

All results are loaded directly from experiment-generated CSV files; the per-run CSVs are immutable and aggregation never overwrites them.

VII. CONCLUSION

We compared four trained autoencoder families — SNN-AE, LSTM-AE, GRU-AE, and a bottlenecked Transformer-AE — under three spike-encoding front ends, against PCA and mean-frame linear references, for 1-D temporal signal

reconstruction on three datasets (FSDD, AudioMNIST, MIT-BIH ECG). The evaluation is leakage-controlled: nested six-fold leave-one-group-out cross-validation with no recording or group crossing a split and SNN hyperparameters selected on the inner validation group only. Differences are reported at the recording level with bootstrap intervals and Holm-corrected tests, with speaker- and seed-level results as robustness evidence; cost is an analytic per-class operation count, explicitly not an energy measurement. This design directly answers the leakage, missing-baseline, missing-significance, and energy-overclaim objections to the earlier version of this study. The harness, profiles, and analysis scripts are released so the protocol can be reused for other 1-D signal families.

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